# Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

## Lesson and revision-note context

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| **Subject** | Mathematics |
| **Grade** | JSS1 |
| **Term** | Second Term |
| **Week** | 9 |
| **Topic** | Simplification of Algebraic Terms |
| **Subtopic** | Simplification of algebraic expressions: order of operations, brackets and word problems |
| **Theme** | — |
| **Curriculum objectives** | • carry out order of operation in algebra<br>• insert/remove bracket and simplify expression<br>• solve word problems involving algebraic expression |
| **Previous knowledge** | • Addition and subtraction of algebraic terms<br>• Multiplication and division of algebraic terms |
| **Lesson sequence** | • carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$<br>• insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$<br>• solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost $h$ naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost $d$ naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad. |
| **Instructional materials** | • Chart showing algebraic expression. |
| **Evaluation** | • Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$<br>• Simplify $7 \times 3a - (3a + 5a) \times 2$ |

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| | • Simplify $12-(7-2f)-(3-f)$<br>• Simplify $(8a-5)+(14-3a)$<br>• A pen and a pencil altogether cost $x$ naira. If a pen cost $y$ naira, find the cost of 3 pencils. |

## Lesson objectives

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| **1.** | carry out order of operation in algebra |
| **2.** | insert/remove bracket and simplify expression |
| **3.** | solve word problems involving algebraic expression |

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

# STUDENT REVISION NOTE

**Topic:** Simplification of algebraic expressions: order of operations, brackets and word problems

**Class:** JSS1 **Subject:** Mathematics **Week:** 9

## Section 1: Order of Operations in Algebra

When you simplify an algebraic expression, you cannot just work from left to right. You must follow the order of operations, which tells you which part to work out first.

The order is:

1. **Brackets** — work out anything inside brackets first.
2. **Order** — powers and roots.
3. **Division and Multiplication** — from left to right.
4. **Addition and Subtraction** — from left to right.

In this section there are no powers, so the order you will use is: brackets first, then division and multiplication (left to right), and then addition and subtraction (left to right).

This is just like working with numbers. For example, in $18 - 6 \times 2$ you cannot do $18 - 6$ first. Multiplication comes before subtraction, so you first work out $6 \times 2 = 12$, then $18 - 12 = 6$. The same rules apply when letters appear.

**Example 1.** Simplify the following.

(a) $3x \times 2 + 5x$

There is no bracket, so multiplication comes first: $3x \times 2 = 6x$. Then $6x + 5x = 11x$.

(b) $17y - 5 \times 2y$

Multiplication first: $5 \times 2y = 10y$. Then $17y - 10y = 7y$.

**Example 2.** Simplify the following.

(a) $7x \div x + 5$

Division first: $7x \div x = \frac{7x}{x} = 7$. Then $7 + 5 = 12$.

(b) $4x - (6 \div 3)$

Brackets first: $6 \div 3 = 2$. Then $4x - 2$. Since $4x$ and $2$ are not like terms, the answer stays $4x - 2$.

(c) $(24x \div 6) + x + (1 \times 5x)$

Work out each bracket: $24x \div 6 = 4x$ and $1 \times 5x = 5x$. So the expression becomes $4x + x + 5x$. Now collect like terms: $4x + x = 5x$, then $5x + 5x = 10x$.

When several operations appear together, tackle the division and multiplication first, always working from left to right, and leave addition and subtraction until the end. Brackets always come before everything else. If a term cannot be combined with another because the letters differ (like $4x$ and $2$), leave it as it is.

#### Practice

1. Simplify $4n + 3n \times 10$.
2. Simplify $32 + 8x \div 2$.
3. Simplify $3 \times 5x + 4x \div 2$.
4. Simplify $36y \div 6 + y + 2 \times 4y$.

## Section 2: Removing Brackets and Simplifying

When you meet a bracket in an algebraic expression, you usually cannot go further until you know what the bracket is doing. Here are the rules for removing a bracket.

1. Always try to simplify the terms inside the bracket first, before removing it.
2. A + sign just before a bracket does not change any of the signs inside. When you drop the bracket, every term comes out exactly as it was.

3. A $-$ sign just before a bracket changes every sign inside. Every $+$ becomes $-$, and every $-$ becomes $+$, when the bracket is removed.

Once the brackets are gone, collect the like terms and simplify as you already know how to.

Example 1. Simplify the following.

(a) $a + (b + c)$

A $+$ sits before the bracket, so the signs inside stay the same:
$$a + (b + c) = a + b + c$$

(b) $a + (b - c)$

Again the $+$ before the bracket leaves everything as it is:
$$a + (b - c) = a + b - c$$

(c) $2a - (3b - c)$

Here a $-$ sits before the bracket, so every sign inside flips. The $-$ in front of $c$ becomes $+$:
$$2a - (3b - c) = 2a - 3b + c$$

Example 2. Simplify the following.

(a) $a + (a - b)$

The $+$ before the bracket keeps the signs, then collect like terms:
$$a + (a - b) = a + a - b = 2a - b$$

(b) $6x + (3 - x)$

Drop the bracket, then collect the $x$ terms together:
$$6x + (3 - x) = 6x + 3 - x = 6x - x + 3 = 5x + 3$$

(c) $(3a + 2) + (5a + a)$

Each bracket is preceded by a $+$, so all signs stay. Drop both brackets, then collect the like terms:
$$(3a + 2) + (5a + a) = 3a + 2 + 5a + a = 3a + 5a + a + 2 = 9a + 2$$

Example 3. Simplify the following.

(a) $5x - (3y + z)$

A $-$ before the bracket flips every sign: the $+3y$ becomes $-3y$ and $+z$ becomes $-z$:
$$5x - (3y + z) = 5x - 3y - z$$

(b) $10d - (9c - 4d)$

Flip every sign inside the bracket, then collect like terms:
$$10d - (9c - 4d) = 10d - 9c + 4d = 10d + 4d - 9c = 14d - 9c$$

(c) $(6x - y) - (4x - 5y)$

The first bracket is dropped as it is; the $-$ before the second bracket flips its signs ($+4x$ becomes $-4x$, $-5y$ becomes $+5y$):
$$(6x - y) - (4x - 5y) = 6x - y - 4x + 5y = 6x - 4x + 5y - y = 2x + 4y = 2(x + 2y)$$

**Practice**

1. Simplify $(12a - 7) + (4 - 5a)$.
2. Simplify $(8x - 2y) - (9x - 4y)$.
3. Simplify $12 - (7 - 2f) - (3 - f)$.
4. Simplify $(8a - 5) + (14 - 3a)$.

## Section 3: Turning Words into Algebra: Word Problems

A word problem describes a situation in sentences, and your job is to turn those words into an algebraic expression. The words tell you the quantity each letter stands for, and they tell you how the quantities are related — with phrases like "less than", "more than", "cost of two things together", or "how many altogether".

Here are the three phrases you will meet most often:

- **"less than"** means subtract. If a pen costs ₦P and a pencil costs 15 naira **less** than a pen, the pencil costs ₦(P − 15).
- **"more than"** means add. If a rubber costs 40 naira **more** than a pencil that costs ₦x, the rubber costs ₦(x + 40).
- **"altogether", "total", "how many now"** mean add everything together.

The key is to write each piece in algebra first, then combine the pieces. Follow these steps:

1. Decide what each letter stands for.
2. Write an expression for each item described in the words.
3. Add or subtract the expressions as the words ask.
4. Simplify by collecting like terms.

**Example 1 — "less than" and "together"**

A pen costs ₦P and a pencil costs 15 naira less than a pen.

(a) Find the cost of a pencil.

The pencil costs 15 naira **less** than the pen, and the pen costs ₦P. So:

cost of a pencil = ₦(P − 15)

(b) Find the cost of a pen and a pencil.

"Together" means add the two costs:

= ₦P + ₦(P − 15)
= ₦P + ₦P − ₦15
= ₦2P − ₦15
= ₦(2P − 15)

So a pen and a pencil together cost ₦(2P − 15).

**Example 2 — "more than" and "together"**

A pencil costs ₦x and a rubber costs 40 naira more than a pencil.

(a) Find the cost of a rubber.

The rubber costs 40 naira **more** than the pencil, and the pencil costs ₦x. So:

cost of a rubber = ₦(x + 40)

(b) Find the cost of a pencil and a rubber.

"Together" means add the two costs:

= ₦x + ₦(x + 40)
= ₦(x + x + 40)
= ₦(2x + 40)

So a pencil and a rubber together cost ₦(2x + 40).

**Example 3 — counting how many altogether**

In a class there are x boys and y girls last semester. This semester, 5 boys join and 3 girls leave the class. How many students are now in this class?

Write each part in algebra first:

number of boys = x + 5 (5 join, so the boys increase)
number of girls = y − 3 (3 leave, so the girls decrease)

Now add them to find the total:

total number = x + 5 + y − 3
= x + y + 5 − 3
= x + y + 2

So there are now (x + y + 2) students in the class.

**Practice**

1. A pen costs ₦h and a crayon costs ₦20 less than a pen.
   (a) Find the cost of a crayon.
   (b) Find the cost of a crayon and a pen.
2. Envelopes cost ₦d per pack, and a notepad costs 20 naira more than a pack of envelopes. Find the total cost of a pack of envelopes and one notepad.

## Check your work

Attempt every question before you read this page.

### Order of Operations in Algebra

1. $34n$

   Multiplication first: $3n \times 10 = 30n$. Then $4n + 30n = 34n$.
2. $32 + 4x$

   Division first: $8x \div 2 = 4x$. Then $32 + 4x$. The terms 32 and $4x$ are not like terms, so the answer stays $32 + 4x$.
3. $17x$

   Work out the multiplication and division first: $3 \times 5x = 15x$ and $4x \div 2 = 2x$. Then $15x + 2x = 17x$.
4. $15y$

   Division and multiplication first: $36y \div 6 = 6y$ and $2 \times 4y = 8y$. The expression becomes $6y + y + 8y$. Collect like terms: $6y + y = 7y$, then $7y + 8y = 15y$.

### Removing Brackets and Simplifying

1. $7a - 3$

   A + before each bracket keeps the signs:
   $$(12a - 7) + (4 - 5a) = 12a - 7 + 4 - 5a = 12a - 5a + 4 - 7 = 7a - 3$$
2. $2y - x$

   Drop the first bracket as it is; the − before the second flips every sign ($+9x$ becomes $-9x$, $-4y$ becomes $+4y$):
   $$(8x - 2y) - (9x - 4y) = 8x - 2y - 9x + 4y = 8x - 9x + 4y - 2y = -x + 2y$$
3. $2 + 3f$

   A − before each bracket flips all the signs inside:
   $$12 - (7 - 2f) - (3 - f) = 12 - 7 + 2f - 3 + f = 12 - 7 - 3 + 2f + f = 2 + 3f$$
4. $5a + 9$

   Each bracket is preceded by a +, so all signs stay:
   $$(8a - 5) + (14 - 3a) = 8a - 5 + 14 - 3a = 8a - 3a + 14 - 5 = 5a + 9$$

### Turning Words into Algebra: Word Problems

1. (a) ₦(h − 20)
   (b) ₦(2h − 20)

   (a) The crayon costs 20 naira less than the pen, so cost of a crayon = ₦(h − 20).
   (b) Crayon and pen together: ₦(h − 20) + ₦h = ₦(h + h − 20) = ₦(2h − 20).
2. ₦(2d + 20)

   A notepad costs 20 naira more than a pack of envelopes, so one notepad = ₦(d + 20).
   Total cost of a pack of envelopes and one notepad = ₦d + ₦(d + 20) = ₦(d + d + 20) = ₦(2d + 20).

## Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

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| **Revision note ID** | cand-539dccb6 | **Date** | |
| **Subtopic** | Simplification of algebraic expressions: order of operations, brackets and word problems | **Rater** | |

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

| **Criterion** (select one score per row) | **1** Poor | **2** Weak | **3** Adequate | **4** Good | **5** Excellent |
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| **Objective and curriculum alignment** Does the revision note cover the intended lesson objectives and curriculum target? | ○ | ○ | ○ | ○ | ○ |
| **Taught-lesson alignment** Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth? | ○ | ○ | ○ | ○ | ○ |
| **Scope and conceptual correctness** Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors? | ○ | ○ | ○ | ○ | ○ |
| **Student-facing clarity and usefulness** Can JSS1 learners understand and use it meaningfully as take-home review material? | ○ | ○ | ○ | ○ | ○ |

| Strengths | Concerns | Recommendation |
|---|---|---|
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## Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

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| ☐ | **Misses a required objective** — Skips something the lesson was meant to teach. |
| ☐ | **Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic). |
| ☐ | **Wrong or misleading content** — Contains a mathematical or factual mistake. |
| ☐ | **Too hard for JSS1 to follow** — Likely confusing for the students. |
| ☐ | **Other:** _____ |